

BSEd Mathematics Student Survey

ExamiQ+ Pilot Study — Pre- and Post-Study Instruments

This survey supports the ExamiQ+ thesis pilot focused on BS Secondary Education Major in Mathematics students. It captures study habits, calibration and metacognition, BSEd-specific learning needs, and feedback for improving the platform beyond basic review. Administer the pre-survey before first use and the post-survey after 2–4 weeks of ExamiQ use.

Unless marked otherwise, use a 1–5 Likert scale: 1 = Strongly disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly agree.

Part A — Pre-Study Survey

Administer before the student's first ExamiQ review session. Items marked 'Core thesis item' align with the pilot protocol for pre/post comparison.

Section 1 — Profile and Context

1. [Multiple choice] What year level are you in?

Note: 1st year | 2nd year | 3rd year | 4th year

2. [Multiple choice] Which math subjects are you currently enrolled in?

Note: Check all that apply (list subjects from current term).

3. [Multiple choice] How many hours per week do you spend reviewing math outside class?

Note: 0–2 | 3–5 | 6–10 | More than 10

4. [Multiple choice] What do you primarily use to review math?

Note: YouTube / modules / classmates / Quizlet / LMS / tutoring / other

Section 2 — Problems Students Experience (Pre-Study)

5. [Likert 1–5] I often feel confident about a math topic but still get it wrong on exams.
6. [Likert 1–5] I sometimes get answers right but do not fully understand why.
7. [Likert 1–5] I find it hard to decide which math topic to study first when preparing for exams.
8. [Likert 1–5] I struggle to explain math solutions step-by-step (not just getting the final answer).
9. [Likert 1–5] I feel unprepared for timed math exams even when I studied the material.
10. [Likert 1–5] I have difficulty connecting math concepts to how I would teach them in a classroom.
11. [Likert 1–5] When I review, I usually re-read notes or watch videos instead of practicing under exam conditions.
12. [Likert 1–5] I rarely know whether my mistakes come from careless errors, formula gaps, or conceptual misunderstanding.
13. [Open-ended] Describe your biggest challenge when reviewing for math exams.
14. [Open-ended] What usually happens when you think you understand a topic but fail the exam?

Section 3 — Calibration and Metacognition (Pre-Study)

15. [Likert 1–5] I am aware of how confidence affects my learning.

Note: Core thesis item (pre).

16. [Likert 1–5] Before answering, I can usually tell whether I truly know a solution or I am guessing.

17. [Likert 1–5] After a wrong answer, I can identify whether I was overconfident or simply did not know the material.

18. [Likert 1–5] Rating or reflecting on confidence after each answer helps me study better.

Note: Core thesis item (pre).

19. [Likert 1–5] I would recommend ExamiQ to classmates.

Note: Core thesis item (pre).

20. [Open-ended] Optional comments before using ExamiQ.

Note: Core thesis open feedback (pre).

Part B — Post-Study Survey

Administer after 2–4 weeks of ExamiQ use. Students may complete this from Profile without a session gate. Retain core thesis items for statistical comparison.

Section 3 — Calibration and Metacognition (Post-Study)

21. [Likert 1–5] ExamiQ helped me understand my calibration gaps.

Note: Core thesis item (post).

22. [Likert 1–5] ExamiQ helped me see when I was overconfident on topics I got wrong.

23. [Likert 1–5] ExamiQ helped me see when I got answers right but was not sure why.

24. [Likert 1–5] ExamiQ helped me decide what to review next more than my usual study methods.

25. [Likert 1–5] ExamiQ's session summary (mastery, misconception, lucky guess, expected gap) made sense to me.

26. [Likert 1–5] Confidence ratings or confidence feedback improved my self-regulated learning.

Note: Core thesis item (post).

Section 4 — BSEd Mathematics-Specific (Post-Study)

27. [Likert 1–5] Practicing math in exam-like timed conditions is important for my success as a future math teacher.

28. [Likert 1–5] Learning to identify common student misconceptions in math would help me in practicum or microteaching.

29. [Likert 1–5] I would find it useful if review feedback showed what type of error I made (conceptual, procedural, or careless).

30. [Likert 1–5] I want review questions aligned to topics I will teach (algebra, geometry, statistics), not only general math.

31. [Likert 1–5] I would use a tool more if it helped me prepare for licensure or board-style math exams, not only classroom quizzes.

32. [Open-ended] As a future math teacher, what do you wish a review system helped you with that YouTube or modules do not?

33. [Open-ended] Which ExamiQ feature was most useful (timed exam, session summary, mistake list, AI tutor, topic recommendations)? Why?

Section 5 — System Improvement and Edge (Post-Study)

34. [Likert 1–5] I would recommend ExamiQ to other BSEd Math students.

Note: Core thesis item (post).

35. [Likert 1–5] ExamiQ is more useful than our LMS quizzes because it shows learning gaps, not only scores.

36. [Likert 1–5] I would use ExamiQ more if my professor assigned review windows tied to our class schedule.

37. [Likert 1–5] I trust AI-generated explanations or tutor chat for math learning.

38. [Ranking] Rank which new features would help you most:

Note: LET or board exam practice mode; error-type labels (misconception categories); 'explain like you are teaching' practice; peer review or study group mode; printable weak-topic report for professor consultation; mobile-friendly offline review

39. [Open-ended] What one feature would make ExamiQ indispensable for BSEd Math students?

40. [Open-ended] What frustrated you or felt missing while using ExamiQ?

Note: Core thesis open feedback (post).

Appendix — Mapping Responses to System Improvements

Use this table when analyzing open-ended answers and Likert patterns. It links common BSEd Math needs to ExamiQ capabilities that provide thesis edge.

If students report...	ExamiQ direction / edge
I do not know what to review first	Strengthen adaptive recommendations and weak-topic reports
I am confident but still wrong	Highlight misconception tier in session summary and dashboard
Timed exams break me	Position timed exam mode as core review, not optional practice
I need to teach, not only solve	Add error-type taxonomy or 'explain the misconception' prompts
Professors notice problems too late	Use faculty intervention export and chair course audit
Questions must match our curriculum	Emphasize BSEd Math program scoping and chair approval workflow
LET prep matters	Pilot board-exam review windows with department-approved items
We do not trust random AI content	Stress human-in-the-loop: faculty create, chair approves